Practice workbook · Network protocols: what they are and which one does what

Why protocols are needed and what named protocols do, including choosing and justifying an email protocol (IMAP/POP3/SMTP) and describing Ethernet.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-03

1. Question 3 · part c · 3 marks

2019 JUN · Paper 01

Worked-example retry · Attempt part c only.

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3 Isra works in an office building where she has access to different types of network.

(a) Isra uses different types of networks for different tasks.

(i) Name the type of network she uses to print a document on the printer in the office next door.

(1)

(ii) Name the type of network she uses to order stationery from an online supplier.

(1)

(b) Identify the measurement used for network data speeds.

(1)

- ☒ **A** Megabits per second
- ☒ **B** Megapixels per second
- ☒ **C** Mebicycles per second
- ☒ **D** Mebihertz per second

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1. Question 3 · part c · 3 marks

2019 JUN · Paper 01

Worked-example retry · Attempt part c only.

(c) Isra uses her tablet computer and smartphone to access email.

She wants to set up a new email account.

State the email protocol she should use.

Justify your choice.

(3)

Email protocol

Justification

(d) The table describes characteristics of different network topologies.

Complete the table to match each characteristic to **one** network topology.

(4)

Characteristic	Bus	Ring	Star	Mesh
Network performance degrades as more devices are added.	☒	☒	☒	☒
All devices are directly connected to all others.	☒	☒	☒	☒
Each node has a physical attachment to a routing device.	☒	☒	☒	☒
There are no data collisions, because packets travel in the same direction.	☒	☒	☒	☒

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1. Question 3 · part c · 3 marks · Working and answer

Working and answer

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2. Question 1 · part f.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part f.i only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

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2. Question 1 · part f.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part f.i only.

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(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

(2)

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

(2)

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2. Question 1 · part f.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part f.i only.

(e) Network layouts can be based upon a number of different network topologies.

Figure 1 shows a network topology.

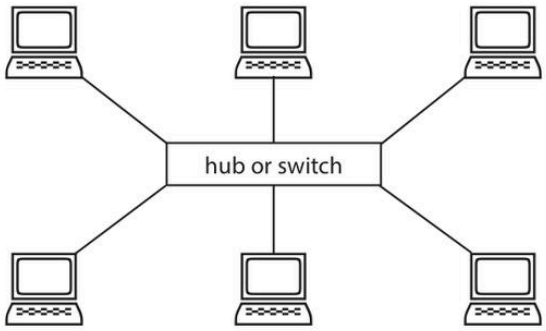


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

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2. Question 1 · part f.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part f.i only.

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(f) The network uses TCP/IP.

Figure 2 shows the TCP/IP protocol stack.

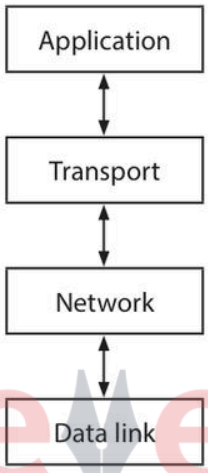


Figure 2

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(i) One use of the application layer is to send and receive emails.

State **two** email protocols.

(2)

1 .....

2 .....

(ii) Upeka sends an email to Peter. The transport layer in Upeka's computer splits the email into data packets.

State **two** tasks performed by the transport layer in Peter's computer when the packets arrive from the network layer.

(2)

1 .....

.....

2 .....

.....

(Total for Question 1 = 17 marks)

2. Question 1 · part f.i · 2 marks · Working and answer

Working and answer

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3. Question 2 · part b · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part b only.

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2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

.....

.....

(b) State the purpose of network protocols.

(1)

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

(1)

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

3. Question 2 · part b · 1 mark · Working and answer

Working and answer

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4. Question 4 · part c.ii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part c.ii only.

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4 Data packets travel across networks from one device to another.

(a) Identify the measurement of network speeds.

(1)

- ☒ **A** Mebibits per second
- ☒ **B** Mebibytes per second
- ☒ **C** Megabits per second
- ☒ **D** Megabytes per second

(b) Data packets contain the addresses of the sender and the receiver.

Complete the table to give the number of bits that make up each type of network address.

(2)

Type	Example	Number of bits
IPv4	192.169.0.3	
IPv6	1050:a500:00c0:0440:0006:0300:700d:436f	

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(c) Data packets travel over physical media.

(i) Describe **one** difference between the media used by a wired network and a wireless network.

(2)

4. Question 4 · part c.ii · 2 marks

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Further practice · Attempt part c.ii only.

(ii) Some people confuse Ethernet® and Wi-Fi.

Describe what is meant by the term **Ethernet®**.

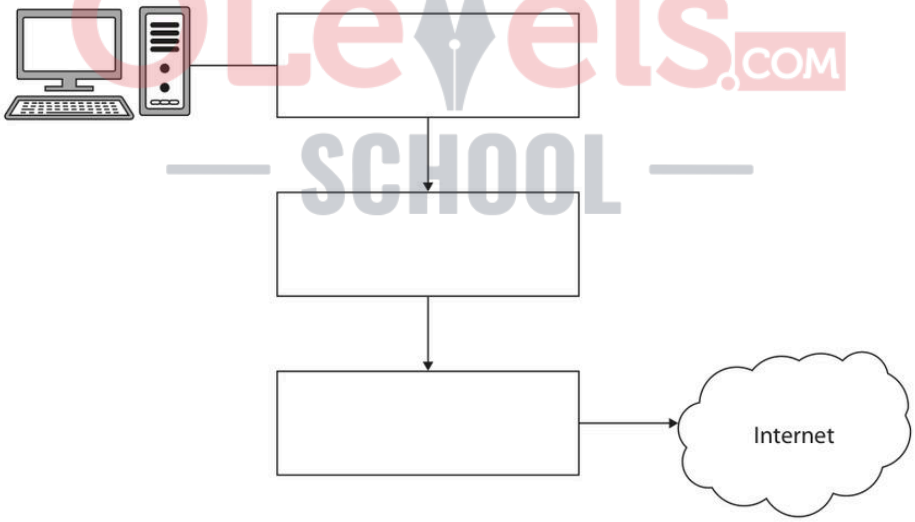
(2)

(d) A single physical box connects a desktop computer to the Internet.

The box incorporates three different components.

Complete the diagram to show the names of the components in the correct order.

(3)



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P 7 2 9 3 7 A 0 1 2 2 4

4. Question 4 · part c.ii · 2 marks · Working and answer

Working and answer

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5. Question 1 · part c · 1 mark

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Further practice · Attempt part c only.

1 Computers can be networked together.

(a) Complete the table by adding a cross ☒ to match **each** statement to **one** networking term.

(4)

Statement	DNS	PAN	WAN	WWW
A network of networks **not** limited to a single location	☒	☒	☒	☒
A network that connects devices within the range of a single user	☒	☒	☒	☒
A set of hypertext documents accessed through the HTTP protocol	☒	☒	☒	☒
A system that maps domain names to IP addresses	☒	☒	☒	☒

(b) Networks use the TCP/IP protocol model.

The application layer and the network layer are two layers in the TCP/IP model.

Give the names of the **other two** layers.

(2)

1 .....

.....

2 .....

.....

(c) Emails can be sent between computers on a network.

Identify the protocol required to **send** an email.

(1)

- ☒ **A** HTTP
- ☒ **B** IMAP
- ☒ **C** POP3
- ☒ **D** SMTP

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2

P 7 8 6 3 7 A 0 2 2 0

5. Question 1 · part c · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part c only.

(d) A library sets up a network using a star topology.

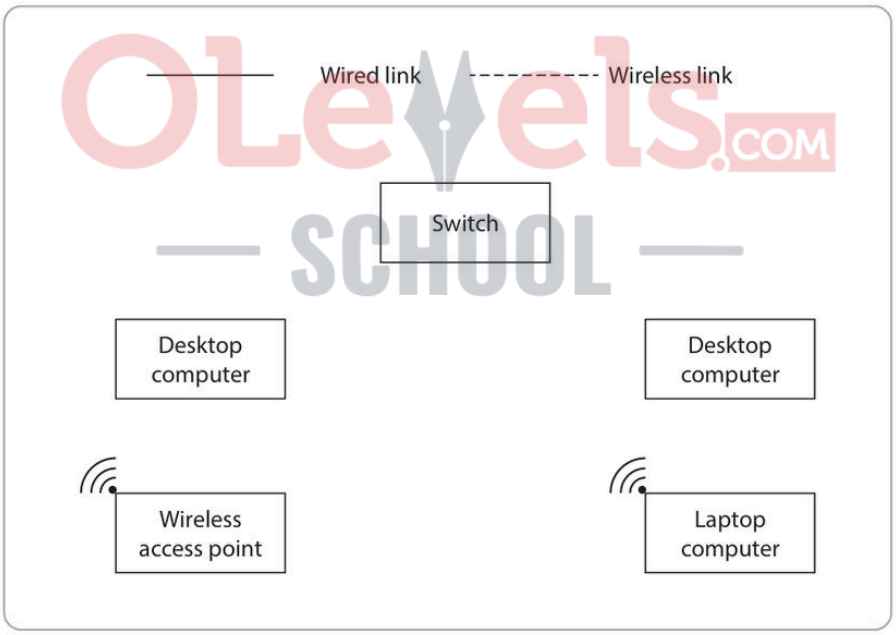
The library is connecting:

- two desktop computers with wired network interface cards
- a laptop with a wireless network interface card
- a switch
- a wireless access point.

Complete the diagram to show how the devices are connected.

Use a solid line to indicate a wired link and a dashed line to indicate a wireless link.

(3)



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5. Question 1 · part c · 1 mark · Working and answer

Working and answer

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